

CSE370 : Database Systems

Assignment 12/11/2025 | Fall 2025

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No 1 Query (as Plain Text)	SELECT C.customer_name, B.loan_number FROM customer C INNER JOIN borrower B ON C.customer_id = B.customer_id INNER JOIN loan L ON B.loan_number = L.loan_number WHERE L.branch_name = 'Downtown';
No 1 SS (of Query & Output in Shell)	<pre>MariaDB [Bank_23101137]> SELECT C.customer_name, B.loan_number -> FROM customer C -> INNER JOIN borrower B ON C.customer_id = B.customer_id -> INNER JOIN loan L ON B.loan_number = L.loan_number -> WHERE L.branch_name = 'Downtown'; +-----+-----+ customer_name loan_number +-----+-----+ Johnson L-14 Jones L-17 Williams L-17 +-----+-----+ 3 rows in set (0.000 sec)</pre>
No 2 Query (as Plain Text)	SELECT C1.customer_name AS customer1, C2.customer_name AS customer2, C1.customer_city AS City from customer C1, customer C2 where C1.customer_city = C2.customer_city AND C1.customer_id > C2.customer_id;

No 2 SS
(of Query & Output
in Shell)

```
MariaDB [Bank_23101137]> SELECT C1.customer_name AS customer1, C2.customer_name AS customer2, C1.customer_city AS City
from customer C1, customer C2 where C1.customer_city = C2.customer_city AND C1.customer_id > C2.customer_id;
+-----+-----+-----+
| customer1 | customer2 | City |
+-----+-----+-----+
| Hayes     | Jones     | Harrison |
| Curry     | Smith     | Rye      |
| Adams     | Lindsay   | Pittsfield |
| Green     | Turner    | Stamford |
+-----+-----+-----+
4 rows in set (0.000 sec)
```

No 3 Query
(as Plain Text)

SELECT B.branch_name AS Branch_name, SUM(A.balance*0.04) AS Total_Interest FROM branch B LEFT JOIN account A ON
B.branch_name = A.branch_name GROUP BY B.branch_name;

No 3 SS
(of Query & Output
in Shell)

```
MariaDB [Bank_23101137]> SELECT B.branch_name AS Branch_name, SUM(A.balance*0.04) AS Total_Interest FROM branch B LEFT JOIN account
A ON B.branch_name = A.branch_name GROUP BY B.branch_name;
+-----+-----+
| Branch_name | Total_Interest |
+-----+-----+
| Brighton    | 66.00         |
| Downtown    | 20.00         |
| Mianus       | 28.00         |
| North Town  | NULL          |
| Perryridge  | 16.00         |
| Pownal      | NULL          |
| Redwood     | 28.00         |
| Round Hill  | 14.00         |
+-----+-----+
8 rows in set (0.000 sec)
```

No 4 Query
(as Plain Text)

SELECT
b.branch_city AS City,
a.account_number,
a.balance
FROM account a

```

INNER JOIN branch b ON a.branch_name = b.branch_name
WHERE a.balance = (
  SELECT MAX(a2.balance)
  FROM account a2
  INNER JOIN branch b2 ON a2.branch_name = b2.branch_name
  WHERE b2.branch_city = b.branch_city
)
ORDER BY b.branch_city, a.account_number;

```

No 4 SS
(of Query & Output
in Shell)

```

MariaDB [Bank_23101137]> SELECT
->   b.branch_city AS City,
->   a.account_number,
->   a.balance
-> FROM account a
-> INNER JOIN branch b ON a.branch_name = b.branch_name
-> WHERE a.balance = (
->   SELECT MAX(a2.balance)
->   FROM account a2
->   INNER JOIN branch b2 ON a2.branch_name = b2.branch_name
->   WHERE b2.branch_city = b.branch_city
-> )
-> ORDER BY b.branch_city, a.account_number;

```

City	account_number	balance
Brooklyn	A-201	900
Horseneck	A-215	700
Palo Alto	A-222	700

3 rows in set (0.001 sec)

**No 5 Query
(as Plain Text)**

```
SELECT *
FROM (
  SELECT
    l.loan_number,
    l.amount AS loan_amount,
    c.customer_name
  FROM loan l
  INNER JOIN borrower b ON l.loan_number = b.loan_number
  INNER JOIN customer c ON b.customer_id = c.customer_id
  ORDER BY l.amount DESC
  LIMIT 5
) AS top5
ORDER BY
  loan_amount ASC,
  loan_number DESC;
```

No 5 SS
(of Query & Output
in Shell)

```
MariaDB [Bank_23101137]> SELECT *
-> FROM (
->     SELECT
->         l.loan_number,
->         l.amount AS loan_amount,
->         c.customer_name
->     FROM loan l
->     INNER JOIN borrower b ON l.loan_number = b.loan_number
->     INNER JOIN customer c ON b.customer_id = c.customer_id
->     ORDER BY l.amount DESC
->     LIMIT 5
-> ) AS top5
-> ORDER BY
->     loan_amount ASC,
->     loan_number DESC;
```

loan_number	loan_amount	customer_name
L-17	1000	Jones
L-16	1300	Adams
L-15	1500	Hayes
L-14	1500	Johnson
L-23	2000	Smith

5 rows in set (0.000 sec)

No 6 Query
(as Plain Text)

```
SELECT DISTINCT c.customer_name
FROM customer c
JOIN depositor d ON c.customer_id = d.customer_id
```

	JOIN borrower b ON c.customer_id = b.customer_id JOIN loan l ON b.loan_number = l.loan_number WHERE l.branch_name = 'Perryridge';
No 6 SS (of Query & Output in Shell)	<pre> MariaDB [Bank_23101137]> SELECT DISTINCT c.customer_name -> FROM customer c -> JOIN depositor d ON c.customer_id = d.customer_id -> JOIN borrower b ON c.customer_id = b.customer_id -> JOIN loan l ON b.loan_number = l.loan_number -> WHERE l.branch_name = 'Perryridge'; +-----+ customer_name +-----+ Hayes +-----+ 1 row in set (0.000 sec) </pre>
No 7 Query (as Plain Text)	SELECT c.customer_name, SUM(l.amount) AS total_loan FROM customer c JOIN borrower b ON c.customer_id = b.customer_id JOIN loan l ON b.loan_number = l.loan_number GROUP BY c.customer_id, c.customer_name HAVING COUNT(*) >= 2;

No 7 SS
(of Query & Output
in Shell)

```
MariaDB [Bank_23101137]> SELECT c.customer_name, SUM(l.amount) AS total_loan  
-> FROM customer c  
-> JOIN borrower b ON c.customer_id = b.customer_id  
-> JOIN loan l ON b.loan_number = l.loan_number  
-> GROUP BY c.customer_id, c.customer_name  
-> HAVING COUNT(*) >= 2;
```

```
+-----+-----+  
| customer_name | total_loan |  
+-----+-----+  
| Smith        |         2900 |  
+-----+-----+  
1 row in set (0.000 sec)
```